

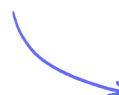
Exam Questions

Trigonometric Graphs & Equations

Trigonometric Graphs / Periods & Amplitudes / Solving Trig Equations

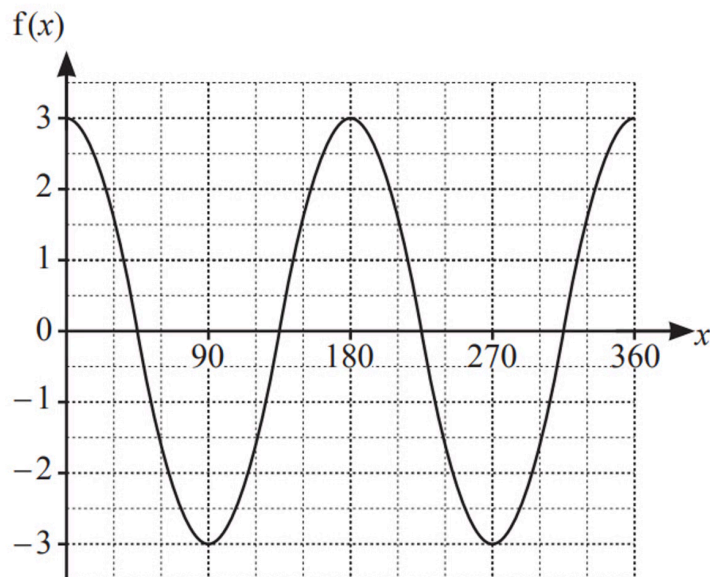
Medium (8 questions)	/20
Hard (7 questions)	/19
Very Hard (7 questions)	/22
Total Marks	/61

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Medium Questions

1 (a)



The graph shows $f(x) = a \cos(bx)^\circ$.

Find the value of a and the value of b .

(2 marks)

(b) Write down the period of $f(x)$.

(1 mark)

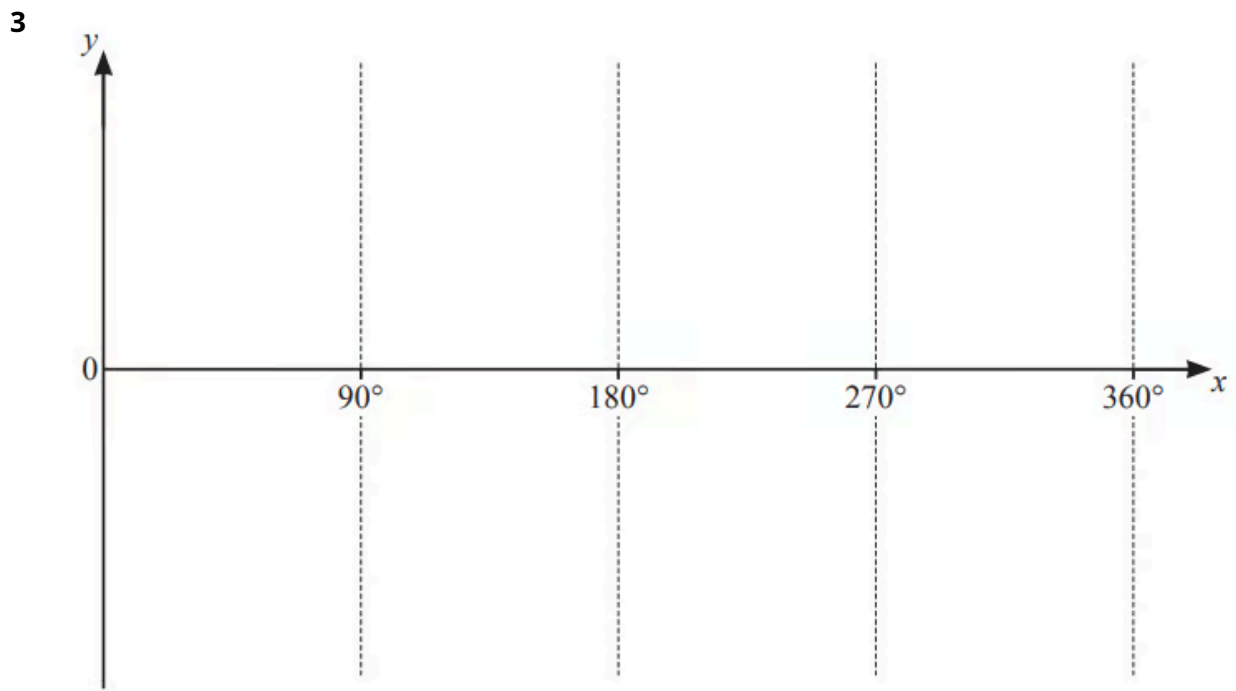
2 $f(x) = 3 \sin(4x^\circ)$

Find the amplitude and period of $f(x)$.

Amplitude =

Period =

(2 marks)



On the diagram, sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$.

(2 marks)

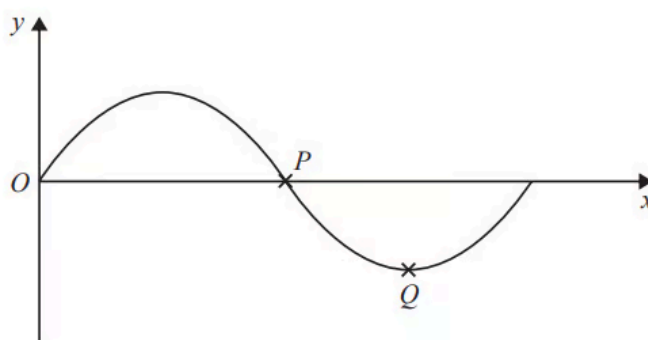
4 (a) When $\sin x^\circ = 0.36$, find the acute angle x° .

(1 mark)

(b) When $\sin x^\circ = 0.36$, find the obtuse angle x° .

(1 mark)

5 (a) The diagram shows part of a sketch of the curve $y = \sin x^\circ$



Write down the coordinates of

- i) the point P
- ii) the point Q

(2 marks)

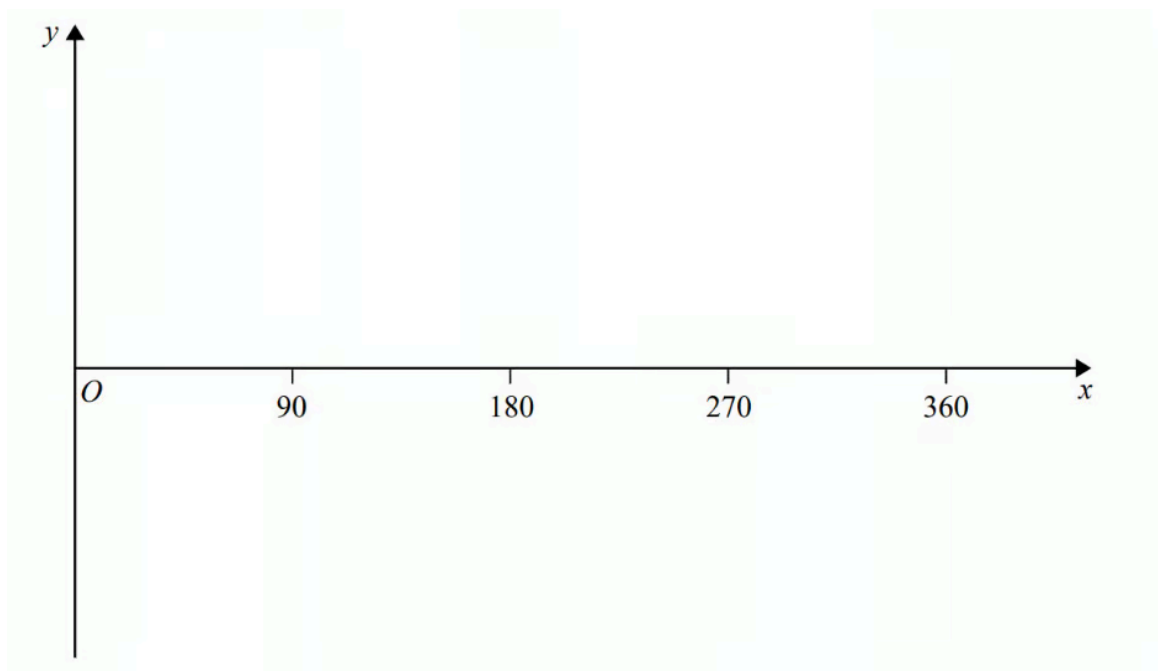
(b) Sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$

Show the coordinates of any points of intersection with the coordinate axes.



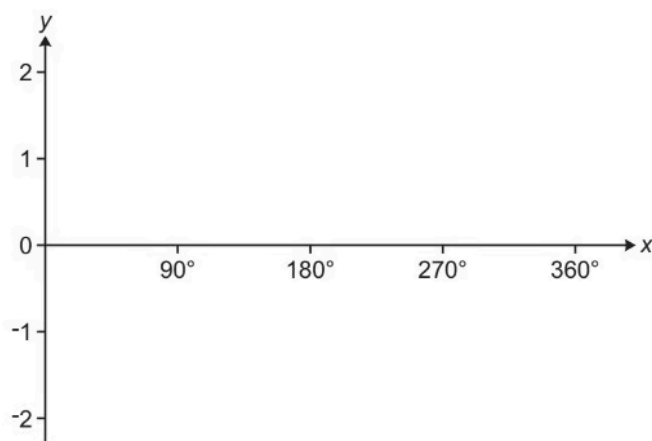
(2 marks)

6 Sketch the graph of $y = \cos x^\circ$ for $0 \leq x \leq 360$



(2 marks)

7 Sketch the graph of $y = \sin x$ for $0^\circ < x < 360^\circ$



(2 marks)

8 $f(x) = p \sin(qx)$

The amplitude of $f(x)$ is 4 and the period is 120° .

Find the exact value of $f(45)$.

(3 marks)

Hard Questions

1 $f(x) = p\cos(qx)$

The amplitude of $f(x)$ is 3 and the period is 30° .

Find $f(10)$.

(3 marks)

2 Solve the equation $\sin x = -0.75$ for $0^\circ \leq x \leq 360^\circ$.

(2 marks)

3 Solve the equation $\tan x = 2$ for $0^\circ \leq x \leq 360^\circ$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(2 marks)

4 Solve $3\tan x = -4$ for $0^\circ \leq x \leq 360^\circ$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(3 marks)

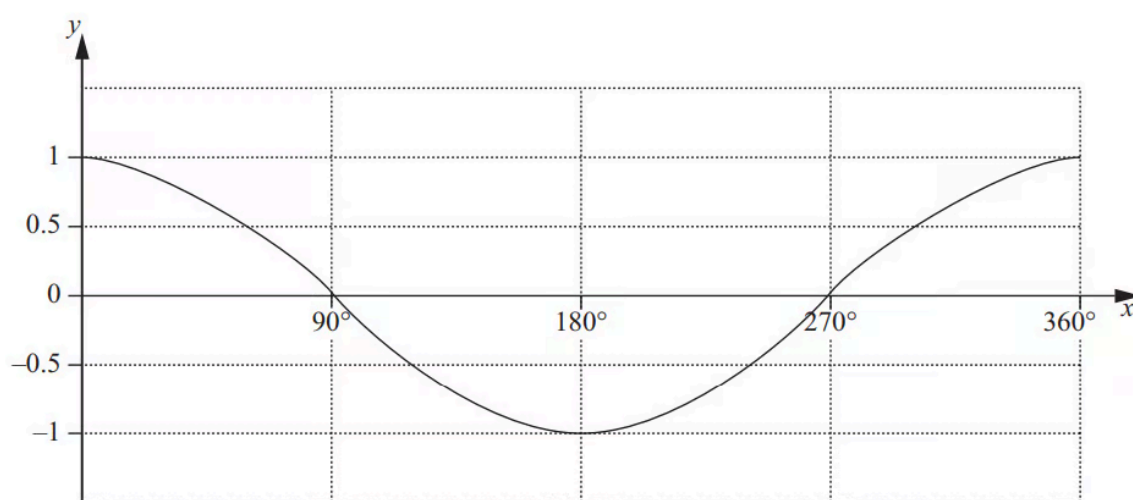
5 x° is an obtuse angle and $\sin x^\circ = 0.43$.

Find the value of x .

$x = \dots\dots\dots$

(2 marks)

6 (a) The grid shows the graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$.



Solve the equation $3\cos x = 1$ for $0^\circ \leq x \leq 360^\circ$.

Give your answers correct to 1 decimal place.

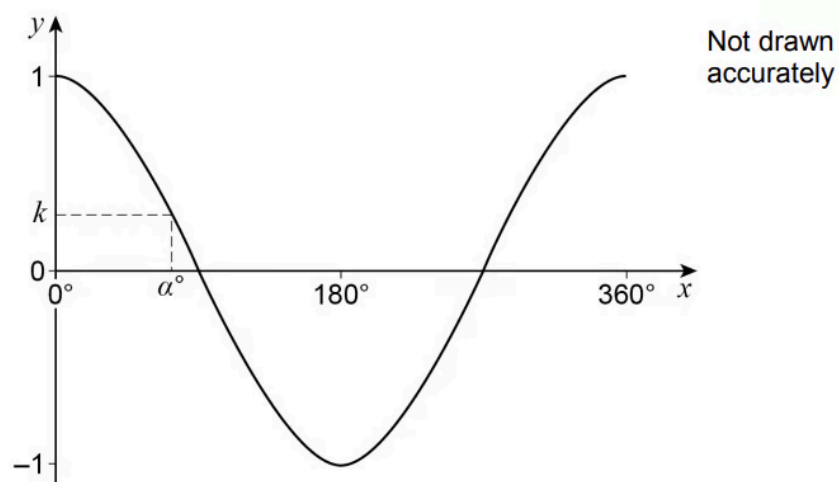
$\dots\dots\dots$ and $\dots\dots\dots$

(4 marks)

(b) On the same grid, sketch the graph of $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$.

(1 mark)

7 (a) Here is a sketch of $y = \cos x$ for values of x from 0° to 360°



α° is an acute angle.

$$\cos \alpha^\circ = k$$

Circle the value of $\cos (180^\circ - \alpha^\circ)$

$1 - k$	k	$-k$	$-1 - k$
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(1 mark)

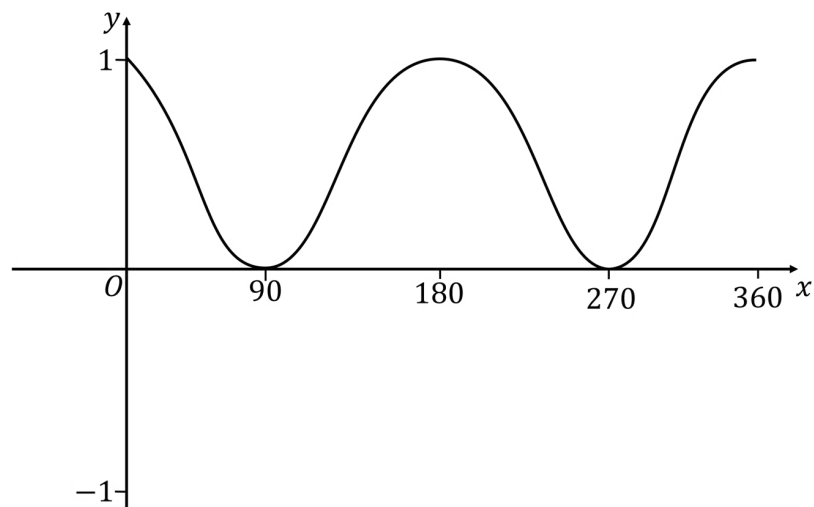
(b) Circle the value of $\cos (360^\circ + \alpha^\circ)$

$k - 1$	$k + 1$	$-k$	k
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(1 mark)

Very Hard Questions

1



The diagram shows the graph of $f(x) = (\cos x^\circ)^2$ for $0 \leq x \leq 360$.

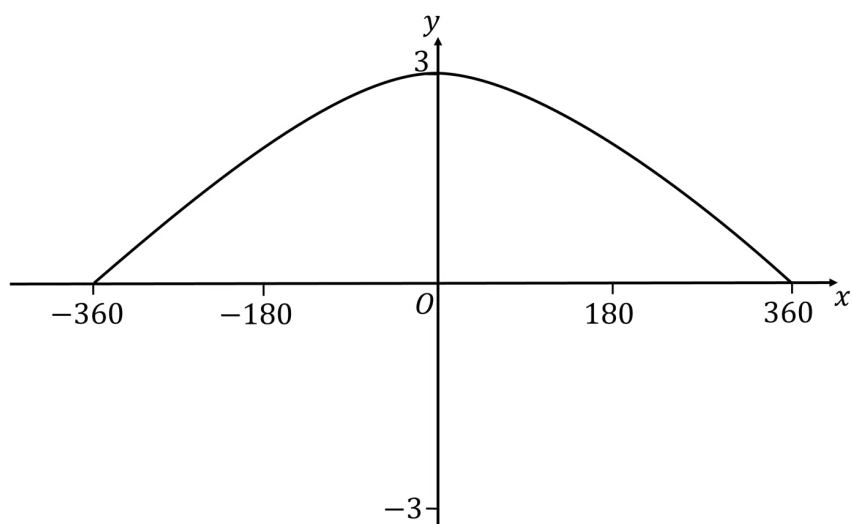
Write down the amplitude and period of $f(x)$.

Amplitude

Period

(2 marks)

2



The diagram shows the graph of $f(x) = a \cos(bx)$ for $0 \leq x \leq 360$.

(i) Write down the amplitude and period of $f(x)$.

Amplitude

Period

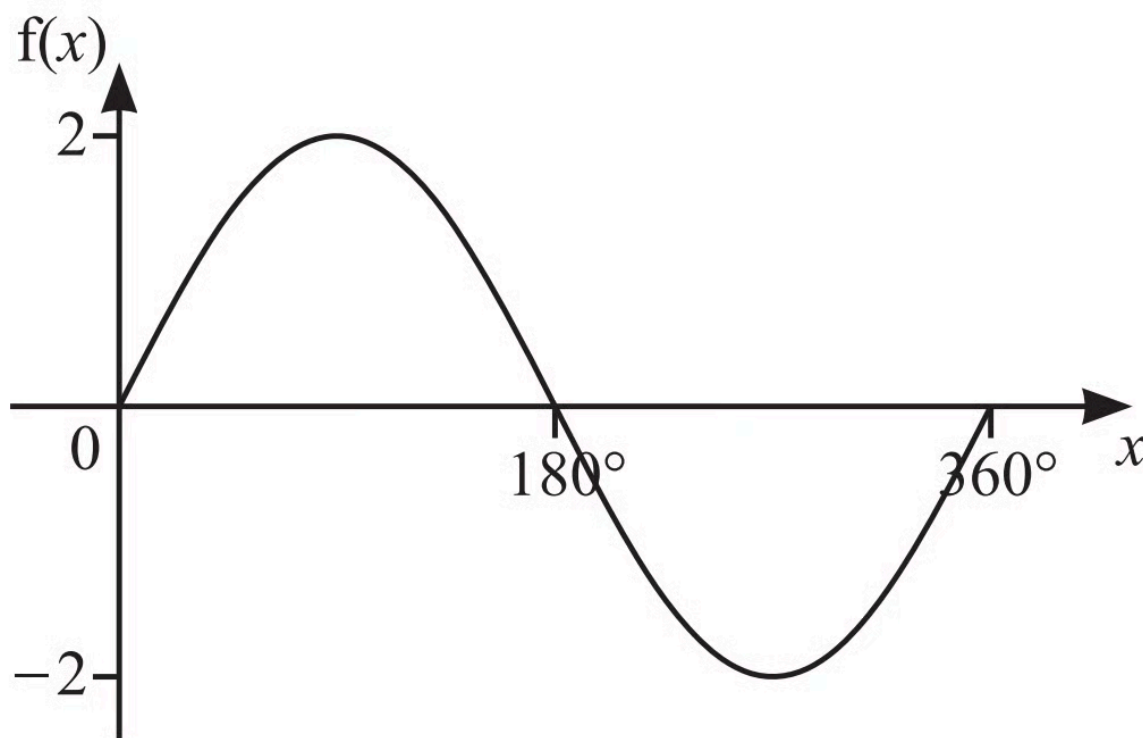
[2]

(ii) Write down the value of a and b .

[1]

(2 marks)

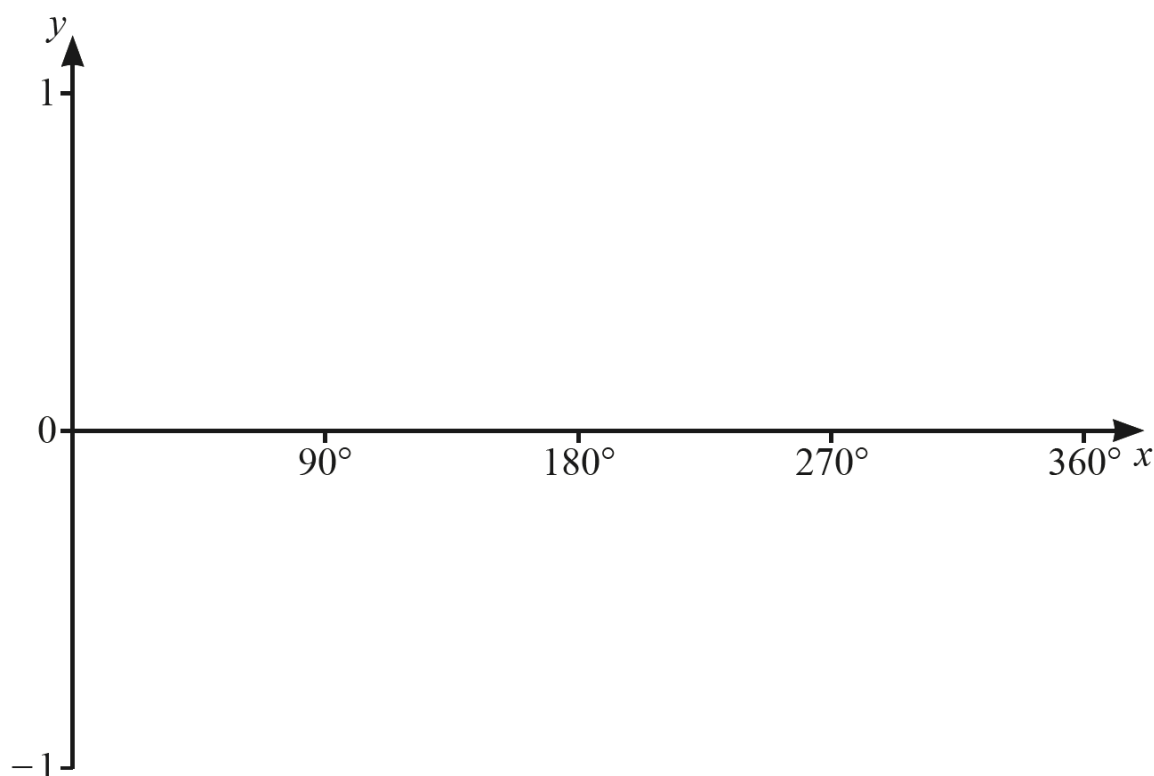
3 The diagrams show the graph of a function. Write down the function.



$f(x) = \dots\dots\dots$

(2 marks)

4 (a) i) On the axes, sketch the graph of $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$.



[2]

ii) Describe fully the symmetry of the graph of $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$.

[2]

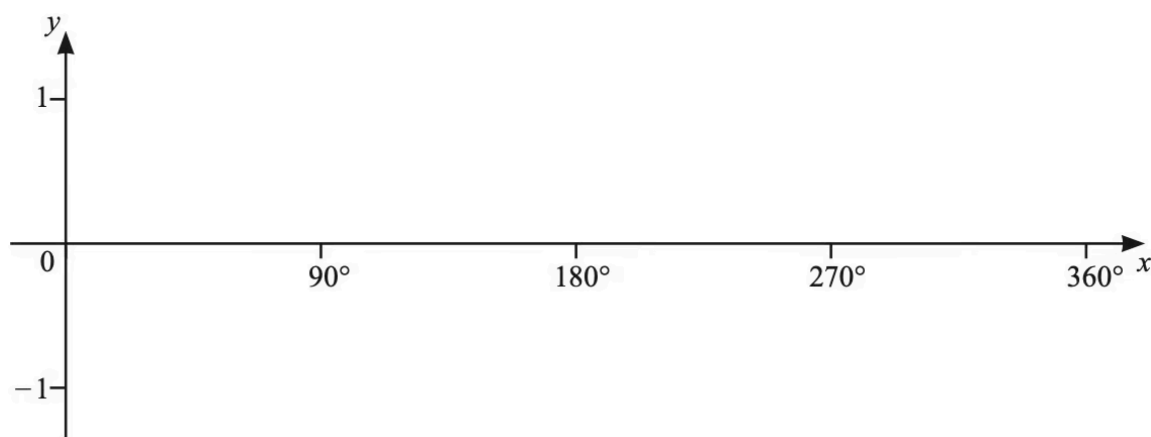
(4 marks)

(b) Solve $4 \sin x - 1 = 2$ for $0^\circ \leq x \leq 360^\circ$.

$x = \dots\dots\dots$ and $x = \dots\dots\dots$

(3 marks)

5 (a)



On the diagram, sketch the graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$.

(2 marks)

(b) Solve the equation $4\cos x + 2 = 3$ for $0^\circ \leq x \leq 360^\circ$.

$x = \dots\dots\dots$ and $x = \dots\dots\dots$

(3 marks)

6 Solve the equation $5 \tan x = -7$ for $0^\circ \leq x \leq 360^\circ$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(3 marks)

7 y is an obtuse angle.

Which statement is true? Tick **one** box.

☐	$\sin y > 0$ and $\cos y > 0$
☐	$\sin y > 0$ and $\cos y < 0$
☐	$\sin y < 0$ and $\cos y > 0$
☐	$\sin y < 0$ and $\cos y < 0$

(1 mark)